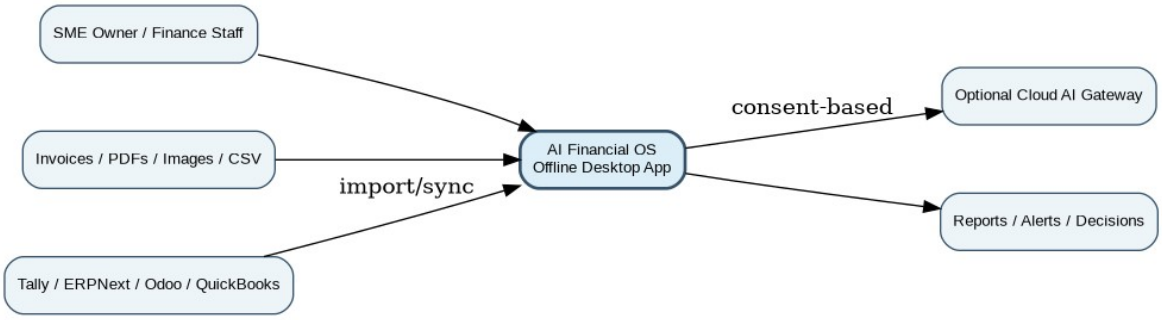
AI Financial Operating System for SMEs

Product Requirements Document and Technical Blueprint



Version 1.0 | 3 August 2026

Prepared as a product-owner reference and an implementation handoff for OpenAI Codex.

Status: Developer-ready planning baseline. This is not legal, tax, accounting, or licensing advice. Validate jurisdiction-specific requirements and all third-party licenses before commercial release.

Document Control

Field	Decision
Document owner	Product founder / product owner
Primary implementation audience	OpenAI Codex, lead engineer, solution architect, QA lead
Product working name	AI Financial Operating System (AIFOS); rename before public launch
Initial market	English-language SMEs; India pilot recommended, followed by UAE and selected export markets
Primary platform	Windows desktop, offline-first
Commercial model	Freemium local core plus optional paid cloud AI decision-support services
Architecture status	Recommended reference architecture; subject to prototype benchmarks and legal review
Explicitly deferred	Blockchain, autonomous payments, full ERP replacement, perfect handwriting recognition, all-language OCR

Change Log

Version	Date	Summary
0.1	Prior discussion	Initial concept: offline OCR, reports, optional AI CFO.
1.0	3 Aug 2026	Full PRD, open-source base strategy, licensing decision gate, architecture, requirements, acceptance criteria, and Codex execution plan.

How to Use This Document

- **For the founder:** Use Sections 1–9 to validate the product and commercial assumptions before development.
- **For Codex:** Use Sections 10–26 as binding implementation requirements. Work milestone-by-milestone; do not build deferred features.
- **For engineers:** Treat requirement IDs and acceptance criteria as the source of truth. Record deviations in an Architecture Decision Record (ADR).
- **For legal/accounting review:** Focus on licensing, privacy, AI disclosures, tax-localization boundaries, and data-processing agreements.
- **For pilots:** Use the success metrics, usability criteria, and pilot plan to evaluate whether the product saves measurable time or prevents errors.

1. Executive Summary

AIFOS is an offline-first desktop application that converts invoices and finance records into a clean, searchable operating view for small and medium-sized businesses. The local product provides document ingestion, English OCR, field extraction, human review, deterministic finance rules, dashboards, alerts, and configurable reports without requiring an internet connection. A paid cloud layer adds conversational analysis, cash-flow scenarios, decision support, and narrative reports through a provider-neutral large-language-model gateway.

The product must not attempt to replace a regulated accountant, independently file taxes, move money, or make irreversible financial decisions. It is an intelligence and workflow layer that improves the quality, speed, and explainability of finance operations while keeping the human owner or accountant in control.

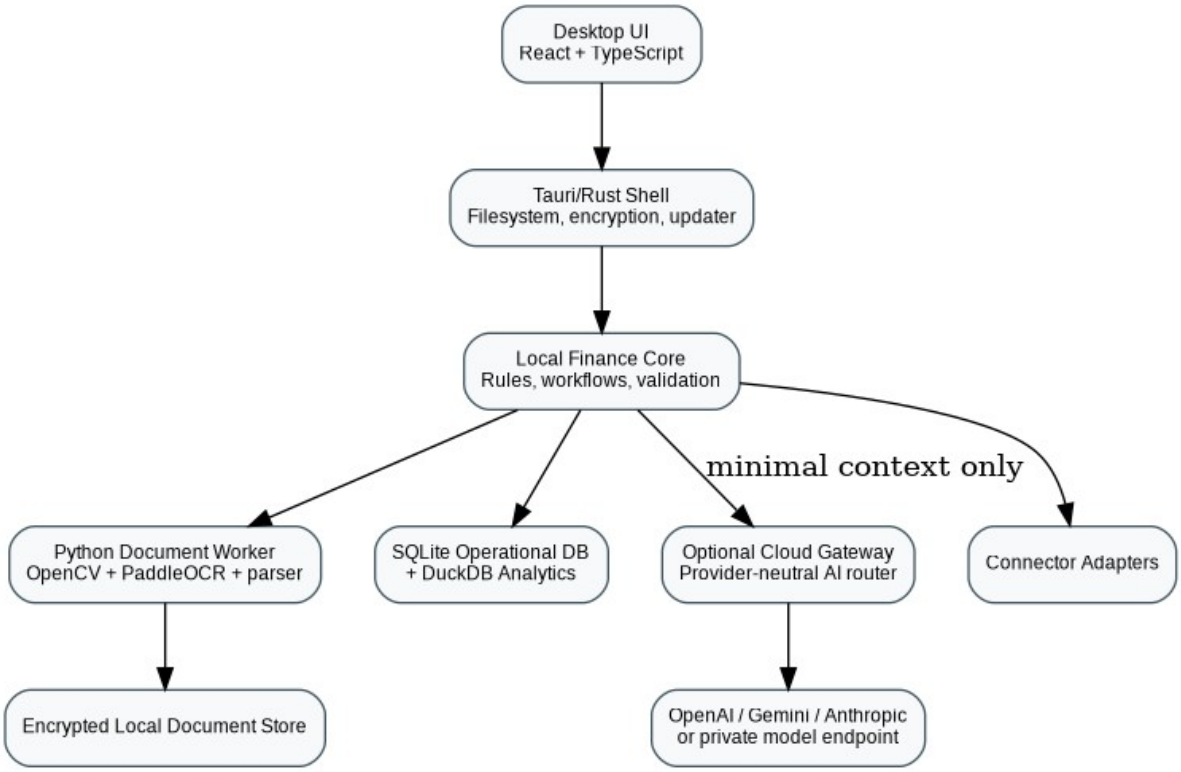


Figure 1 — Recommended reference architecture

1.1 Executive Product Decisions

ID	Decision	Rationale
ED-01	Build a lightweight offline desktop product first.	Target hardware and privacy needs make a browser-only ERP unsuitable as the primary experience.
ED-02	No local 8B LLM in MVP.	Most target PCs lack RAM/GPU capacity; deterministic rules and OCR create most of the early value.
ED-03	Use cloud AI only after explicit user action.	Keeps the base product usable offline and makes privacy boundaries understandable.
ED-04	Do not build a complete accounting ledger initially.	Import from existing systems or maintain a staging/subledger sufficient for intelligence and reports.
ED-05	Blockchain is deferred.	It adds no essential value to the first single-company invoice workflow.
ED-06	Choose the open-source base only after a license strategy decision.	Frappe Books is the fastest open-core route but is AGPL-3.0; a permissive custom core

ID	Decision	Rationale
		offers greater proprietary flexibility.
ED-07	English printed invoices first.	Handwriting and multiple languages are later capabilities driven by pilot demand.
ED-08	Every recommendation must be explainable.	The user must see the data, assumptions, and rules supporting an alert or AI answer.

1.2 Recommended Base Strategy

Recommended commercial architecture: Build the proprietary/lightweight desktop core with Tauri, React/TypeScript, SQLite, DuckDB, and a separate Python OCR worker. Integrate with ERPNext, Odoo, Tally, QuickBooks, and Frappe Books through adapters instead of embedding their full codebases.

For a rapid market-validation prototype, a separate branch may fork Frappe Books. That route can save significant UI and accounting groundwork, but the modified desktop application must be managed under the requirements of its AGPL-3.0 license. The cloud AI service should remain a cleanly separated service with an independently reviewed interface. Obtain specialist open-source counsel before choosing that route for a commercial product.

2. Product Definition

2.1 Vision

Give every SME a privacy-conscious finance operations assistant that turns messy documents and disconnected records into timely, understandable, and actionable information—without forcing the business to replace the tools it already trusts.

2.2 Product Promise

- Import invoices in minutes, not hours.
- See what is due, unusual, missing, duplicated, or becoming risky.
- Generate useful reports offline.
- Ask deeper finance questions only when the user chooses cloud AI.
- Keep source documents, extracted fields, and corrections auditable.
- Work alongside accountants and existing accounting software rather than claiming to replace them.

2.3 Problem Statement

Many SMEs receive invoices through email, WhatsApp, paper, scanners, supplier portals, and accounting exports. Staff manually read them, rename files, retype fields, reconcile totals, decide which supplier to pay, and prepare recurring reports. Traditional accounting applications are systems of record; they often do not provide a unified document-intelligence and decision-support layer across these unstructured inputs. The result is clerical effort, delayed visibility, duplicate or incorrect entries, missed due dates, and decisions based on incomplete data.

2.4 Jobs to Be Done

ID	Job
JTBD-01	When invoices arrive in mixed formats, I want them captured and organized automatically so that staff do not retype every field.
JTBD-02	When cash is limited, I want a transparent priority view so that I understand which obligations are urgent and why.
JTBD-03	When a supplier changes prices, I want an early alert so that I

ID	Job
	can negotiate or source alternatives.
JTBD-04	When my accountant asks for records, I want a complete, traceable package so that audit preparation is faster.
JTBD-05	When I review performance, I want customized reports in my business language so that the numbers lead to decisions.
JTBD-06	When I ask a complex question, I want an AI answer grounded in my own transactions, with citations and assumptions.
JTBD-07	When the internet is unavailable or sensitive data must stay local, I want the core product to continue working.

3. Target Users and Markets

3.1 Primary Personas

Persona	Context	Pain	Primary value
Owner-manager	10–100 employee manufacturer/trader	Cash position, overdue receivables, supplier dues, margin changes	Dashboard, priority list, plain-language reports
Finance executive	Processes documents and reconciles entries	Manual typing, duplicate invoices, missing fields, month-end pressure	OCR queue, validation, exports, exception handling
Accountant / advisor	Reviews books and compliance	Inconsistent source documents, unclear audit trail	Reviewer access, correction history, standard reports
Operations / purchase manager	Controls suppliers and materials	Price drift, quantity mismatch, delivery/invoice differences	Supplier trends, PO/GRN matching later
External auditor / consultant	Needs evidence without editing data	Time-consuming document requests	Read-only audit pack and evidence links

3.2 Market Sequence

Stage	Market	Scope
Pilot	India, English-first SMEs	Manufacturing/trading businesses already using Tally, spreadsheets, or PDFs.
Expansion 1	UAE SMEs	English and Arabic document roadmap; VAT-oriented reports; privacy and data-hosting review.
Expansion 2	US/Europe niche verticals	QuickBooks/Xero connectors, jurisdiction packs, GDPR/privacy requirements, industry-specific workflows.

Market discipline: Do not launch globally with generic claims. Prove one vertical and one workflow: invoice ingestion → review → due-date/price alerts → offline management report.

4. Product Principles, Scope, and Boundaries

4.1 Product Principles

- **Offline by default:** No login or internet connection should be required for core document and report workflows.
- **Human-verifiable intelligence:** Every extracted value and recommendation links to source evidence.
- **Rules before generative AI:** Use deterministic calculations for totals, aging, thresholds, and compliance dates.

- **Progressive enhancement:** Cloud AI adds value but does not hold the base product hostage.
- **Integration over replacement:** Read from and export to existing accounting/ERP systems.
- **Low-end hardware awareness:** Avoid background model processes and excessive memory use.
- **Safe failure:** Low-confidence documents go to review; the system never silently invents financial facts.
- **Modular architecture:** OCR engines, model providers, and ERP connectors must be replaceable.

4.2 MVP In Scope

- Windows desktop installer and local company workspace.
- English printed invoice import from PDF, JPG, PNG, and folder watch.
- Local OCR, extraction, confidence scoring, and review queue.
- Vendor/customer master and invoice staging records.
- Duplicate detection, price-change alerts, due-date alerts, aging, basic anomaly rules.
- Offline dashboards and configurable reports.
- CSV/Excel-compatible import/export and PDF report export.
- Optional cloud AI analysis with explicit consent and grounded answers.
- Encrypted local backups and restore.
- Basic ERP connector framework; one pilot connector selected after discovery.

4.3 Explicitly Out of Scope for MVP

- Full double-entry accounting replacement or statutory books of account.
- Automated tax filing, audit opinion, or legal/accounting advice.
- Autonomous bank payments or payment initiation.
- Perfect handwriting recognition.
- All Indian languages, Arabic, or global tax-localization on day one.
- Blockchain-based invoice transfer, tokens, cryptocurrency, or smart contracts.
- Payroll, CRM, manufacturing planning, and full inventory ERP.
- Continuous cloud synchronization of all raw documents.
- Fully autonomous AI agents negotiating or committing to supplier terms.

5. Open-Source Base Platform Decision

5.1 Decision Gate

Before Codex writes production code, the founder must choose one of the following license strategies. This is a commercial and legal decision, not only a technical one.

Option	Base	Advantages / constraints	Use when
A. Custom permissive core (recommended)	Tauri + React + SQLite + DuckDB + Python OCR	More initial development; maximum control; lightweight; easier proprietary licensing	Best for long-term commercial product
B. Frappe Books fork	Fork existing offline accounting desktop app	Fastest prototype; substantial accounting/UI base; AGPL-3.0 obligations	Best only if open-core distribution is acceptable
C. ERPNext/Frappe web app	Build an ERPNext app or external integration	Rich ERP; server-heavy; ERPNext GPL-3.0; not ideal for old offline PCs	Best for enterprise/web pilots
D. Odoo Community module	Add intelligence as Odoo modules/services	Large ecosystem; web/server deployment; LGPL core with edition/module complexity	Best when customer already uses Odoo

5.2 Recommended Decision

Use Option A for the production architecture. Build a narrow desktop intelligence product rather than a full ledger. Use Frappe Books as a reference implementation and optional import target. Run a time-boxed Option B prototype only if the team accepts an open-core licensing model and receives legal approval.

5.3 Why Frappe Books Matters

Frappe Books is the most directly relevant open-source base discovered for this concept: it is an offline desktop accounting application using Electron, Vue, and SQLite, with invoicing, payments, double-entry accounting, dashboards, and financial reports. Its repository includes development and build instructions. However, it is licensed under AGPL-3.0, so a commercial fork must be planned around source-disclosure and distribution obligations. [OS-01]

5.4 License Rules for the Project

- Create a THIRD_PARTY_NOTICES.md and software bill of materials (SBOM) from the first commit.
- Pin dependency versions and preserve copyright/license notices.
- Do not copy source code from GPL/AGPL projects into a proprietary core without approved legal strategy.
- Keep GPL/AGPL integrations process-separated or API-based where practical, but do not assume separation automatically removes obligations.
- Check transitive dependencies; a permissive top-level license does not guarantee every optional dependency is suitable.
- Have counsel review model-weight licenses independently from inference-runtime licenses.
- Document every fork, patch, and upstream commit hash used.

6. Verified Open-Source Components and Download Instructions

The following list was checked against official repositories on 3 August 2026. “Recommended” means technically suitable for evaluation, not a legal approval.

ID	Project	Role	License	Official repository	Start command
OS-01	Frappe Books	Optional rapid prototype/reference	AGPL-3.0	https://github.com/frappe/books	git clone https://github.com/frappe/books.git
OS-02	ERPNext	ERP integration/reference	GPL-3.0	https://github.com/frappe/erpnext	git clone https://github.com/frappe/erpnext.git
OS-03	Frappe Framework	Optional web/admin framework	MIT	https://github.com/frappe/frappe	git clone https://github.com/frappe/frappe.git
OS-04	Odoo Community	Optional ERP integration	LGPL-3.0	https://github.com/odoo/odoo	git clone --branch 19.0 https://github.com/odoo/odoo.git
OS-05	Tauri	Recommended desktop shell	MIT / Apache-2.0	https://github.com/tauri-apps/tauri	npm create tauri-app@latest
OS-06	PaddleOCR	Primary OCR candidate	Apache-2.0	https://github.com/PaddlePaddle/PaddleOCR	pip install paddleocr
OS-07	Tesseract	OCR fallback/benchmark	Apache-2.0	https://github.com/tesseract-ocr/tesseract	Install OS package; keep language files versioned
OS-08	Docling	Document parsing benchmark	MIT (models vary)	https://github.com/docling-project/docling	pip install docling
OS-09	SQLite	Operational local database	Public domain	https://sqlite.org/	Bundled through application library
OS-10	DuckDB	Local analytics engine	MIT	https://github.com/duckdb/duckdb	Use official binding; pin version
OS-11	Apache ECharts	Dashboard visualizations	Apache-2.0	https://github.com/apache/echarts	npm install echarts
OS-12	Ollama	Optional developer/local-model	Runtime/repo license must be rechecked;	https://github.com/ollama/ollama	Use only in advanced/dev mode;

ID	Project	Role	License	Official repository	Start command
		runtime	model licenses vary		not MVP dependency
OS-13	Qwen open-weight models	Optional future local/cloud model	Model-specific; current Qwen3 open weights described as Apache-2.0	https://github.com/QwenLM/Qwen3	Do not bundle until model license and hardware profile are approved
OS-14	LangChain	Optional provider/tool adapter	MIT	https://github.com/langchain-ai/langchain	Prefer thin internal adapters; add only if it reduces code
OS-15	LlamaIndex	Optional document/RAG framework	MIT	https://github.com/run-llama/llama_index	Use selectively; avoid unnecessary abstraction
OS-16	Qdrant	Future vector search service	Apache-2.0	https://github.com/qdrant/qdrant	Not required for MVP; SQLite FTS may suffice
OS-17	Chroma	Prototype vector store	Apache-2.0	https://github.com/chroma-core/chroma	Prototype only; not required for core

6.1 Frappe Books Evaluation Commands

```
git clone https://github.com/frappe/books.git
cd books
# Follow the repository's current Node.js prerequisite
# Install the package manager specified by the repo
# Then:
yarn
yarn dev
# Build installer:
yarn build
```

Do not begin by modifying the fork: First run the unmodified application, map its data model and extension points, and document which modules would be reused. Make the license decision before committing proprietary business logic to the fork.

6.2 Recommended New Project Scaffold

```
# Desktop shell
npm create tauri-app@latest aifos-desktop

# Suggested choices
# Frontend: React + TypeScript + Vite
# Package manager: pnpm

cd aifos-desktop
pnpm install
pnpm tauri dev

# Separate Python document worker
python -m venv .venv
# Windows: .venv\Scripts\activate
pip install fastapi uvicorn pydantic pillow opencv-python-headless paddleocr
```

7. Core User Journeys

UJ-01 First-time setup

1. Install desktop app.
2. Create local company workspace.
3. Choose currency, financial year, and date format.
4. Set local backup location.
5. Optionally import vendors/customers and historical invoices.
6. No cloud account is required for the free offline mode.

UJ-02 Import and review invoice

7. Drag files, scan, or monitor an import folder.
8. System validates file type and creates immutable source hash.
9. OCR and parsing run locally.
10. User sees extracted fields beside source image.
11. Low-confidence fields are highlighted.
12. User corrects and approves.
13. Approved record enters staging and analytics.

UJ-03 Monitor obligations

14. Dashboard groups overdue, due soon, and later obligations.
15. Rule engine calculates urgency and penalties from entered terms.
16. User can filter by supplier, amount, date, category, and confidence.
17. No payment is executed by the system.

UJ-04 Create an offline report

18. Choose template, date range, filters, and grouping.
19. Preview tables and charts.
20. Add/remove sections.
21. Export PDF/CSV without internet.

UJ-05 Ask cloud AI

22. User presses Ask AI on a report or dashboard.
23. System displays what data will leave the device.
24. User confirms.
25. Context builder selects and redacts minimum necessary data.
26. AI returns an answer with transaction/report citations and assumptions.
27. User can save the answer locally or discard it.

UJ-06 Accountant review

28. Owner exports an audit pack or creates read-only local reviewer access.
29. Reviewer sees source document, extracted values, corrections, and rule events.
30. Corrections are logged with timestamp and user.

8. Functional Requirements

8.1 Workspace, Identity, and Onboarding

ID	Priority	Requirement	Acceptance
FR-ONB-001	Must	Create one or more local company workspaces with independent databases and document stores.	A user can create, open, rename, back up, and restore a company without internet.
FR-ONB-002	Must	Allow optional local PIN/password protection and OS-keychain integration.	The workspace cannot be opened without the configured credential; recovery warnings are clear.
FR-ONB-003	Must	Configure base currency, locale, date format, financial year, tax labels, and report branding.	Settings are reflected in dashboards and exports.
FR-ONB-004	Should	Import vendor/customer masters from CSV.	Validation errors are reported per row and do not block valid rows.
FR-ONB-005	Should	Create role profiles: Owner, Finance Operator, Reviewer.	Permissions are enforced locally and included in the audit log.

8.2 Document Ingestion

ID	Priority	Requirement	Acceptance
FR-ING-001	Must	Import PDF, JPG, JPEG, PNG, and supported image-based scans through drag/drop and file picker.	Each file receives an import ID, SHA-256 hash, original filename, and timestamp.
FR-ING-002	Must	Support batch import with progress, pause, resume, and per-file failure status.	One malformed file does not fail the batch.
FR-ING-003	Must	Provide a configurable watched folder.	New files are copied into the managed store only after stable-size checks.
FR-ING-004	Must	Detect exact duplicate files by hash and probable duplicate invoices by normalized fields.	User can keep, link, or reject duplicates.
FR-ING-005	Should	Import structured CSV exports from existing accounting systems.	Mapping templates can be saved and reused.
FR-ING-006	Could	Capture documents from email or mobile companion.	Deferred until cloud/privacy model is validated.

8.3 OCR and Document Understanding

ID	Priority	Requirement	Acceptance
FR-OCR-001	Must	Preprocess images locally: orientation, deskew, denoise, contrast, crop, and page segmentation.	The original is preserved; processed derivatives are reproducible.
FR-OCR-002	Must	Run English printed-text OCR locally without a GPU.	OCR works on the minimum supported PC within the performance budget.
FR-OCR-003	Must	Extract invoice number, invoice date, supplier, tax identifier, currency, subtotal, tax, total, due date, payment terms, line items when reliable, and bank details as untrusted text.	Every field includes confidence and source coordinates.
FR-OCR-004	Must	Show side-by-side source and extracted fields with click-to-highlight bounding boxes.	A user can verify where every value came from.
FR-OCR-005	Must	Route low-confidence or mathematically inconsistent documents to review.	No low-confidence record is auto-approved.
FR-OCR-006	Must	Validate totals: subtotal + tax ±	Inconsistencies generate an

ID	Priority	Requirement	Acceptance
		adjustments = total within configurable tolerance.	explicit exception.
FR-OCR-007	Should	Learn reusable supplier templates from confirmed documents without training a general model.	Template rules are editable and reversible.
FR-OCR-008	Could	Add Hindi/Gujarati/Arabic printed OCR packs.	Only after language-specific benchmark corpus and customer demand exist.
FR-OCR-009	Won't MVP	Guarantee handwriting recognition.	Handwritten documents are marked unsupported or sent to manual entry.



Figure 2 — Local OCR and review pipeline

8.4 Review and Correction Workflow

ID	Priority	Requirement	Acceptance
FR-REV-001	Must	Maintain statuses: Imported, Processing, Needs Review, Approved, Rejected, Exported.	Status transitions follow a documented state machine.
FR-REV-002	Must	Record original OCR value, corrected value, user, time, and reason.	Audit history is immutable through the UI.
FR-REV-003	Must	Support keyboard-first review for high-volume operators.	A trained user can validate common fields without mouse-only interaction.
FR-REV-004	Should	Surface repeated correction patterns and propose a vendor template update.	Proposal requires user approval and has a rollback option.
FR-REV-005	Should	Allow comments and internal tags.	Comments do not modify accounting values.

8.5 Vendor, Customer, and Category Intelligence

ID	Priority	Requirement	Acceptance
FR-MST-001	Must	Maintain normalized vendor/customer records with aliases and identifiers.	Duplicate entities can be merged with a reversible audit event.
FR-MST-002	Must	Show invoice count, spend/revenue, average payment terms, price trends, and exception rate per entity.	Metrics reconcile to approved records.
FR-MST-003	Should	Classify invoices and line items into editable categories.	User corrections take priority over automated suggestions.
FR-MST-004	Should	Flag new or changed bank details for manual verification.	The product never treats OCR-extracted bank details as trusted payment instructions.
FR-MST-005	Could	Calculate supplier reliability using delivery/PO data.	Requires future PO/GRN integration.

8.6 Offline Rule Engine

Rule ID	Name	Logic	Output
RULE-001	Exact duplicate file	Same content hash	Block or link import
RULE-002	Probable duplicate invoice	Same normalized vendor + invoice number; or high similarity amount/date	Review alert
RULE-003	Arithmetic mismatch	Extracted components do not	Needs Review

Rule ID	Name	Logic	Output
		reconcile to total	
RULE-004	Price increase	Unit price or normalized invoice category increases above configurable threshold versus baseline	Dashboard alert
RULE-005	Due soon	Due date within configured window	Priority queue
RULE-006	Overdue	Due date passed and unpaid/uncleared	Escalated alert
RULE-007	Penalty exposure	Payment terms include structured late fee entered/confirmed by user	Estimated cost with assumptions
RULE-008	Unusual amount	Amount outside configurable historical range	Review alert
RULE-009	New bank details	Bank field differs from confirmed vendor master	High-severity manual verification
RULE-010	Missing tax identifier	Required field missing under selected jurisdiction pack	Compliance warning, not legal conclusion
RULE-011	Cash gap	Projected obligations exceed entered available cash in a future bucket	Scenario warning
RULE-012	Concentration risk	Spend or receivable exposure to one party exceeds threshold	Management alert

8.7 Payment Priority View

The payment priority view is a decision-support queue, not an autonomous payment system. It must use explicit factors and allow the owner to override them.

Factor	Definition	Illustrative weight
Due date urgency	Days until/past due	0–30
Confirmed penalty/interest	Cost of delay based on entered terms	0–20
Operational criticality	User-defined supplier/category criticality	0–20
Discount opportunity	Confirmed early-payment discount	0–10
Relationship risk	User-defined strategic importance	0–10
Amount/cash fit	Scenario-dependent affordability	0–10
Data confidence	Reduce score when source data is uncertain	-20–0

Explainability requirement: The queue must show the score components, not only a final ranking. The user can edit weights and save named strategies such as “Avoid penalties” or “Protect production.”

8.8 Dashboard

ID	Priority	Requirement	Acceptance
FR-DAS-001	Must	Overview cards: payable due, overdue, receivable due, cash entered/imported, invoices needing review, duplicates, and exceptions.	Each card opens the underlying filtered records.
FR-DAS-002	Must	Charts: aging buckets, due-date calendar, supplier spend trend, category trend, and price-change alerts.	Charts are available offline and exportable.
FR-DAS-003	Must	Support period, vendor/customer, category, status, and confidence filters.	Filters persist per workspace.
FR-DAS-004	Should	Allow widget reorder and saved dashboard views.	User can reset to default.
FR-DAS-005	Should	Show data freshness and source coverage.	The user sees which records are imported and which systems are

ID	Priority	Requirement	Acceptance
			not connected.

8.9 Offline Reporting

ID	Report	Minimum content
RPT-001	Payables aging	Open/payable staging by aging bucket, vendor, due date, amount, and confidence.
RPT-002	Receivables aging	Imported sales invoices/receivables by customer and bucket.
RPT-003	Invoice register	Approved and rejected documents with source links and extraction status.
RPT-004	Supplier price trend	Unit price or normalized category spend over time.
RPT-005	Exception report	Duplicates, arithmetic mismatches, missing fields, changed bank details, unusual amounts.
RPT-006	Cash obligation calendar	Expected outflows by day/week/month with scenario assumptions.
RPT-007	Vendor summary	Spend, frequency, payment terms, exceptions, and price movement.
RPT-008	Audit pack	Selected records, source hashes, corrections, approvals, and attached source documents.
RPT-009	Custom table report	User-selected fields, filters, grouping, sort, totals, and saved template.
RPT-010	Management pack	Configurable 4–5 page offline PDF with cover, KPIs, charts, tables, and notes.

8.10 Custom Report Builder

ID	Priority	Requirement	Acceptance
FR-RPT-001	Must	Choose dataset, columns, filters, grouping, sort, totals, chart, title, and branding.	A non-technical user can create and save a report template.
FR-RPT-002	Must	Preview before export.	Preview matches exported pagination within reasonable limits.
FR-RPT-003	Must	Export PDF and CSV; optionally XLSX after dependency review.	Exports contain generation time and source period.
FR-RPT-004	Should	Schedule generation locally while app is running.	No cloud scheduler is required for free mode.
FR-RPT-005	Should	Add text notes and management commentary manually.	Notes are labeled as user-entered.
FR-RPT-006	Should	Expose “Ask AI about this report” in premium mode.	AI context is limited to report data and cited records.

8.11 Cloud AI CFO

The cloud AI feature is an advisory interface over structured, user-approved data. It must never be allowed to fabricate balances, change records silently, or execute transactions.

ID	Priority	Requirement	Acceptance
FR-AI-001	Must	Require explicit consent for each cloud analysis or a clearly revocable session consent.	The UI shows categories and approximate record count being sent.
FR-AI-002	Must	Use a provider-neutral backend adapter.	Provider can be changed by configuration without changing desktop UI.
FR-AI-003	Must	Ground answers in selected structured records and reports.	Every material numeric claim has a citation/link to local source rows.
FR-AI-004	Must	Return structured JSON plus rendered narrative.	Invalid schema results are retried or shown as an error, not silently accepted.

ID	Priority	Requirement	Acceptance
FR-AI-005	Must	Show assumptions, confidence, and a “not accounting advice” disclosure.	Disclosure is visible in every saved AI report.
FR-AI-006	Must	Support questions on profit movement, payment priorities, overdue exposure, supplier trends, and cash scenarios.	Benchmark questions meet accuracy thresholds on test datasets.
FR-AI-007	Should	Generate management commentary for an existing offline report.	Commentary cannot alter report figures.
FR-AI-008	Should	Support model/provider selection by admin policy.	End users see a simple mode, not model complexity.
FR-AI-009	Could	Bring-your-own API key mode.	Only after secure key storage and support implications are addressed.
FR-AI-010	Won’t MVP	Autonomous negotiation, emails, or payments.	AI outputs drafts only, if added later.

8.12 Integrations

ID	System	Stage	Requirement
INT-001	CSV/Excel-compatible files	MVP	Import masters and transactions; export approved records and reports.
INT-002	Tally	Pilot candidate	Use documented export/import or approved interfaces; discovery required for customer versions and schemas.
INT-003	ERPNext	Post-MVP	REST/API adapter; map suppliers, invoices, payments, and attachments.
INT-004	Odoo	Post-MVP	API adapter for customers already on Odoo.
INT-005	QuickBooks Online	Expansion	OAuth cloud connector; region and plan constraints require validation.
INT-006	Xero/Zoho Books	Expansion	Provider adapters based on pilot demand.
INT-007	Email/WhatsApp ingestion	Later	High privacy and platform-policy complexity; not part of initial build.

8.13 Backup, Restore, and Portability

ID	Priority	Requirement	Acceptance
FR-BKP-001	Must	Create encrypted full backup containing database, documents, configuration, and audit log.	Restore on a clean installation reproduces the workspace.
FR-BKP-002	Must	Support manual backup and configurable reminders.	User sees last successful backup and destination.
FR-BKP-003	Must	Export all structured data in documented CSV/JSON formats.	Customer is not locked into an unreadable proprietary database.
FR-BKP-004	Should	Support incremental backups.	Failure cannot corrupt the last valid full backup.
FR-BKP-005	Should	Verify backup integrity without restoring.	Verification result and timestamp are logged.

9. Offline and Online Feature Matrix

Feature	Offline	Internet required	Notes
Workspace/company setup	Yes	No	All configuration local.

Feature	Offline	Internet required	Notes
Invoice/PDF/image import	Yes	No	Source retained locally.
English OCR	Yes	No	CPU-optimized; no local LLM.
Review and corrections	Yes	No	Full audit history.
Duplicate/mismatch/price rules	Yes	No	Deterministic.
Dashboards	Yes	No	ECharts over local queries.
Custom report builder	Yes	No	PDF/CSV export.
Basic payment priority	Yes	No	Transparent configurable scoring.
Narrative explanation of report	No	Yes	Premium AI.
Open-ended CFO questions	No	Yes	Grounded data only.
Advanced cash-flow scenario narrative	Partial	Yes	Calculations local; narrative cloud.
Multilingual cloud document fallback	No	Yes	Future opt-in, only with consent.
Subscription/license validation	Graceful cached mode	Occasional	Must not block access to local data.
Software updates	No	Yes	Manual/offline installer also available.
Telemetry	Off by default	Optional	Explicit opt-in only.

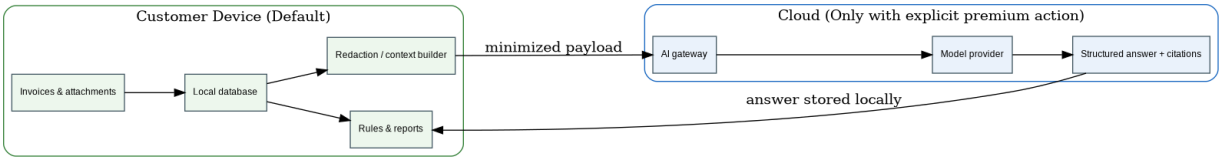


Figure 3 — Privacy boundary between local core and optional cloud AI

10. Reporting Specification

10.1 Offline Management Pack (4–5 Pages)

Page	Content
Page 1 — Executive overview	Company, period, data coverage, payable/receivable totals, overdue totals, documents needing review, and major alerts.
Page 2 — Cash and obligations	Weekly obligation chart, top upcoming payments, overdue items, and user-entered cash scenario.
Page 3 — Supplier/customer movements	Top counterparties, price increases, concentration, and aging.
Page 4 — Exceptions and controls	Duplicates, arithmetic mismatches, changed bank details, missing fields, and review backlog.
Page 5 — Notes and action list	User-entered notes plus rule-generated action checklist; no generative narrative required.

10.2 Premium AI Management Report

- Uses the same locked numeric tables as the offline report.
- Adds plain-language explanations of major changes.
- Identifies contributing records with citations.
- Presents scenarios, not predictions stated as facts.
- Lists assumptions and missing data.
- Ends with a proposed action list requiring user approval.
- Labels all AI-generated text and provider/model execution metadata in an appendix.

10.3 Report Data Contract

```
{
 "report_id": "uuid",
 "workspace_id": "uuid",
 "template_id": "management_pack_v1",
 "period": {"from": "YYYY-MM-DD", "to": "YYYY-MM-DD"},
 "currency": "INR",
 "filters": {},
 "metrics": [{"key": "payables_due", "value": 0, "source_query_id": "Q-101"}],
 "tables": [{"id": "upcoming_payments", "rows": []}],
 "charts": [{"id": "aging", "series": []}],
 "data_quality": {"coverage": 0.0, "review_pending": 0},
 "generated_at": "ISO-8601",
 "generator": "offline-report-engine"
}
```

11. Technical Architecture

11.1 Recommended Technology Stack

Layer	Choice	Reason
Desktop shell	Tauri 2.x / Rust	Small footprint, OS integration, secure capability model, cross-platform future.
Frontend	React + TypeScript + Vite	Large ecosystem, Codex-friendly, maintainable component architecture.
UI components	Accessible headless components + custom design system	Avoid locking core UX to a proprietary component suite.
Operational DB	SQLite	Single-file local transactional database; mature and portable.
Analytics	DuckDB	Fast local analytical SQL over SQLite exports/Parquet where useful.
Charts	Apache ECharts	Offline interactive charts and export support.
Document worker	Python 3.12+ sidecar/service	Best ecosystem for OCR and document processing; version to be benchmarked.
OCR	PaddleOCR primary, Tesseract benchmark/fallback	CPU-capable open-source options; language packs modular.
Document parsing	Custom invoice parser; benchmark Docling	Avoid heavy dependencies unless benchmark proves value.
Local search	SQLite FTS5	Sufficient before adopting a vector database.
Cloud API	FastAPI or equivalent containerized service	Provider abstraction, auth, metering, policy, and structured AI output.
Cloud DB	PostgreSQL	Accounts/subscriptions/configuration only; raw customer documents not stored by default.
Object storage	Optional encrypted object storage	Only for explicit cloud workflows or support uploads.
CI/CD	GitHub Actions	Build, tests, SBOM, license scan, signed release workflow.
Marketing/docs	Vercel or Netlify	Static/Next.js site; separate from sensitive AI gateway.
Cloud runtime	Managed container service	Better for controlled egress, private networking, and background jobs than static hosting.

11.2 Process Boundaries

- Tauri main process owns filesystem capabilities, encryption keys, updates, and process lifecycle.
- Frontend cannot access arbitrary filesystem paths; it calls explicit Tauri commands.
- Python OCR worker receives a job ID and controlled file path, then returns structured JSON.
- Database writes occur through a local application service, not directly from UI components.
- Cloud AI requests pass through a context builder and policy engine before network transmission.
- ERP connectors are isolated adapters with mapping tests and idempotency keys.

11.3 Suggested Monorepo Structure

```
aifos/
 apps/
  desktop/              # React + Tauri app
  cloud-gateway/        # Optional premium AI API
  docs-site/            # Marketing and documentation
 services/
  document-worker/      # Python OCR and parsing
 packages/
  domain/               # Finance entities and calculations
  rules/                # Deterministic rule engine
  report-engine/        # Query/report definitions
  ai-contracts/         # JSON schemas and prompts
  connector-sdk/        # ERP adapter interfaces
  ui/                   # Shared UI components
 connectors/
  csv/
  erpnext/
  odoo/
  tally/                # after discovery
 schemas/
  sqlite/
  api/
 fixtures/
  invoices/
  expected/
 docs/
  adr/
  security/
  licensing/
 scripts/
.github/workflows/
LICENSE
THIRD_PARTY_NOTICES.md
SECURITY.md
CONTRIBUTING.md
```

README.md

12. OCR and Document Intelligence Design

12.1 Processing Stages

Stage	Behavior
1. Intake	Copy source to managed store; compute hash; identify MIME type; reject executables and unsupported archives.
2. Rasterization	Convert PDF pages to images using a license-approved library; preserve original.
3. Preprocessing	Rotate, deskew, denoise, normalize contrast, and detect page boundaries.
4. OCR	Run English model; return text, boxes, and confidence.
5. Layout parsing	Detect header fields, totals, tables, and footer/payment terms.
6. Entity resolution	Match supplier/customer aliases against local master.
7. Normalization	Dates, currency, decimal separators, tax identifiers, and amount precision.
8. Validation	Arithmetic, duplicate, expected-field, and anomaly checks.
9. Confidence policy	Approve only when policy thresholds and validations pass; otherwise review.
10. Feedback	Store corrections for templates/rules; do not train global models on customer data without consent.

12.2 Confidence Policy

Class	Condition	Action
High	≥ configured threshold and validations pass	Eligible for one-click approval; never silently posted to external ERP.
Medium	Some fields uncertain or layout unfamiliar	Needs Review with uncertain fields highlighted.
Low	Core fields missing, handwriting, poor scan, or mismatch	Manual entry/rejection; offer rescanning guidance.

12.3 OCR Benchmark Plan

- Create a legally usable corpus of at least 500 representative English invoices across 30+ suppliers.
- Include born-digital PDFs, scans, phone photographs, skew, stamps, faint print, tables, and multi-page invoices.
- Measure field-level precision/recall, exact-match rate, total arithmetic accuracy, processing time, and memory peak.
- Compare PaddleOCR, Tesseract, and Docling-assisted pipelines on the same corpus.
- Choose the lightest pipeline meeting target accuracy; retain a pluggable engine interface.
- Never report overall OCR character accuracy alone; measure the financial fields that matter.

12.4 Target OCR Metrics for Pilot

Metric	Target
Invoice number exact match	≥ 95% on supported printed invoices
Invoice date exact match	≥ 97%
Supplier match	≥ 95% after alias configuration
Total amount exact match	≥ 98%
Tax amount exact match	≥ 95% where present
Due date/terms	≥ 90% or route to review
False auto-approval of wrong total	< 0.1%

Metric	Target
Median single-page processing on recommended PC	≤ 8 seconds
Peak memory per worker	≤ 1.5 GB target

13. Data Architecture and Domain Model

13.1 Core Entities

Entity	Purpose
Workspace	Local company, settings, encryption metadata.
User	Local user/role and authentication metadata.
Party	Vendor or customer with aliases and identifiers.
Document	Immutable source metadata, hash, pages, storage path.
ExtractionRun	OCR engine/version, timing, text/boxes, confidence.
InvoiceDraft	Normalized extracted/staged invoice fields.
InvoiceLine	Description, quantity, unit, rate, tax, total, category.
ReviewEvent	Correction, approval, rejection, comment, reason.
PaymentTerm	Structured due date, discount, penalty terms confirmed by user.
RuleDefinition	Versioned deterministic rule configuration.
RuleEvent	Triggered rule with inputs, output, severity, and status.
ReportTemplate	Saved offline report definition.
ReportRun	Frozen result dataset and export metadata.
AIRequest	Consent, selected data, redaction summary, provider, model, cost.
AIResponse	Structured answer, citations, assumptions, safety metadata.
ConnectorConfig	External system mapping and sync state.
SyncEvent	Import/export idempotency and error audit.
AuditEvent	Security-sensitive action log.

13.2 Key Data Rules

- Source documents are append-only; replacement creates a new document linked to the old one.
- Extracted values are never overwritten without a ReviewEvent.
- Approved invoice snapshots are versioned.
- All money uses integer minor units or fixed-precision decimals; never binary floating point.
- Every calculated metric stores currency and period.
- AI text is not a ledger value and cannot mutate approved records.
- Deleting a workspace requires explicit confirmation and offers backup first.
- Connector imports use external IDs and idempotency keys to prevent duplication.

13.3 Illustrative Invoice Schema

```
{
 "invoice_id": "uuid",
 "document_id": "uuid",
 "party_id": "uuid|null",
 "invoice_number": {"value": "ABC-123", "confidence": 0.98, "bbox": [0,0,0,0]},
 "invoice_date": {"value": "2026-08-01", "confidence": 0.99},
 "due_date": {"value": "2026-08-31", "confidence": 0.82, "confirmed": false},
 "currency": "INR",
 "subtotal_minor": 100000,
 "tax_minor": 18000,
```

```
  "total_minor": 118000,
  "status": "needs_review",
  "validation": {"arithmetic_ok": true, "duplicate_score": 0.04},
  "version": 1
}
```

14. Deterministic Intelligence and Forecasting

14.1 Rule Engine Design

- Rules are versioned JSON/TypeScript definitions with explicit input fields, thresholds, output severity, and explanation templates.
- Every RuleEvent stores the exact rule version and input snapshot used.
- Rules can be enabled/disabled and thresholds changed per workspace.
- Built-in rules are protected from silent modification; user overrides create a new version.
- Rules operate only on approved or explicitly included draft data.
- Rule evaluation is repeatable: same input + same rule version = same output.

14.2 Cash Scenario Engine

The offline engine should calculate scenarios using known/entered dates and amounts. It must distinguish committed obligations, expected items, and assumptions. The AI may explain scenarios but must not perform the arithmetic independently when a deterministic calculation is available.

- Opening cash/bank balance entered or imported by the user.
- Approved payable/receivable dates and amounts.
- User-defined expected collections and confidence percentages.
- Recurring obligations entered as rules.
- Payment strategy and reserve threshold.
- What-if adjustments such as delayed collection, early payment, or new purchase.

15. Cloud AI System Design

15.1 Model Strategy

Do not hard-code the product to one LLM vendor. Implement an internal model interface with adapters. Start with one provider to reduce complexity, but require contract tests that allow a second provider to be added without changing domain code.

Task	Policy
Narrative explanation	Summarize a frozen report; no recalculation.
Question answering	Answer using retrieved local records and computed metrics.
Scenario communication	Explain deterministic scenario outputs and trade-offs.
Classification fallback	Optional categorization where rules are uncertain; user confirmation required.
Drafting	Draft internal action lists or supplier questions; never send automatically.

15.2 Context Minimization

- Prefer aggregates and selected rows over raw full databases.
- Remove names, bank details, tax identifiers, and addresses unless essential to the question.
- Never send source images by default.

- Show the user a preflight summary of fields and record count.
- Apply per-tenant retention policy at the gateway and avoid storing provider payloads unless required for debugging and consented.
- Use request IDs and local citations instead of revealing unnecessary file paths or personal information.

15.3 Structured AI Response Contract

```
{
 "answer": "string",
 "summary_points": ["string"],
 "numeric_claims": [
  {"claim": "string", "value": 0, "currency": "INR", "citation_ids": ["row:..."]}
 ],
 "recommendations": [
  {"action": "string", "reason": "string", "priority": "low|medium|high", "requires_human_approval": true}
 ],
 "assumptions": ["string"],
 "missing_data": ["string"],
 "confidence": "low|medium|high",
 "disclaimer": "Decision support only; verify with your accountant or advisor."
}
```

15.4 AI Guardrails

- Reject prompts asking the model to fabricate invoices, hide transactions, evade taxes, or falsify evidence.
- Do not provide a definitive legal/tax conclusion; identify that professional review may be required.
- Never expose system prompts, other customers' data, API keys, or internal secrets.
- Treat invoice text as untrusted content; defend against prompt injection inside documents.
- Validate JSON schema and citations before displaying a "high confidence" answer.
- Cap context size, output size, retries, and spend per workspace.
- Provide a clear delete/export control for AI history.

15.5 Cost Controls

- Cache report-level aggregates locally and avoid resending unchanged context.
- Use a small/low-cost model for classification and a stronger model only for complex analysis.
- Summarize long transaction sets deterministically before model calls.
- Budget per plan and display fair-use limits rather than per-question token complexity.
- Log provider cost metadata separately from customer finance data.
- Alert administrators when a prompt pattern causes repeated retries or unusually high spend.

16. Interfaces and API Contracts

16.1 Local Document Worker API

```
POST /v1/jobs/ocr
{
 "job_id": "uuid",
 "file_path_token": "opaque-token",
 "language": "en",
```

```
 "engine": "paddleocr",
 "options": {"extract_tables": true}
}

GET /v1/jobs/{job_id}
→ {"status":"queued|running|completed|failed", "progress":0.0, "result":{...}}
```

16.2 Cloud AI Gateway API

```
POST /v1/analysis
Authorization: Bearer <workspace-session-token>
Idempotency-Key: <uuid>
{
 "task": "explain_report|answer_question|scenario_commentary",
 "question": "string|null",
 "report_contract": {...},
 "selected_records": [...],
 "redaction_summary": {...},
 "response_schema_version": "1.0"
}
```

16.3 Connector Interface

```
interface FinanceConnector {
 id: string;
 capabilities(): ConnectorCapabilities;
 testConnection(config: SafeConfig): Promise<TestResult>;
 importParties(cursor?: string): AsyncIterable<PartyBatch>;
 importInvoices(cursor?: string): AsyncIterable<InvoiceBatch>;
 exportApprovedInvoices(batch: ExportBatch): Promise<ExportResult>;
 getSyncCheckpoint(): Promise<Checkpoint>;
}
```

17. UX and Information Architecture

17.1 Primary Navigation

Area	Purpose
Home	KPIs, due/overdue, review backlog, major alerts.
Inbox	Imported documents and processing state.
Review	Side-by-side source validation.
Invoices	Approved/staged invoice register.
Parties	Vendors/customers and intelligence.
Payments	Priority and obligation calendar; no payment execution.
Reports	Templates, custom builder, exports, Ask AI.
AI Advisor	Premium conversation history and saved analyses.
Integrations	Imports, exports, connector health.
Settings	Company, rules, security, backups, subscription, privacy.

17.2 UX Requirements

- Core workflows must be keyboard accessible and readable at 125% Windows scaling.
- Financial amounts use locale-aware formatting but store normalized values.
- Warnings use text and icons, not color alone.
- No dark-pattern upgrade prompts; premium actions are clearly labeled before use.
- Every AI answer provides “View supporting data.”
- Every destructive action offers confirmation; high-impact actions support undo where feasible.
- Error messages state what failed, what data was preserved, and the next action.
- Processing queues remain usable while background OCR runs.

17.3 Key Screen Acceptance

Screen	Minimum content
Invoice review	Source on left, fields on right, confidence highlights, validation banner, Save/Approve/Reject, keyboard shortcuts.
Dashboard	Loads under performance target, no misleading zeroes when data is missing, cards link to records.
Custom report	Field chooser, filters, grouping, chart options, preview, save template, export.
Ask AI preflight	Question, selected scope, data-to-be-sent summary, estimated premium usage, consent.
AI answer	Narrative, cited figures, assumptions, missing data, actions, disclaimer, save/export.

18. Security, Privacy, and Compliance Requirements

18.1 Threat Model Summary

Threat	Risk	Controls
Lost/stolen laptop	Local financial documents exposed	Workspace encryption, OS credential integration, auto-lock, backup encryption.
Malicious invoice/PDF	Parser exploit or prompt injection	File-type validation, sandboxed worker, patched libraries, treat document text as untrusted.
Compromised update channel	Malicious release	Signed installers, verified update manifests, rollback, reproducible build target.
Cloud credential leak	Unauthorized AI usage/data access	Server-side secrets, key rotation, scoped tokens, no provider keys in desktop app.
Cross-tenant cloud leak	Customer data disclosure	Strict tenant isolation, minimal cloud storage, access tests, audit logs.
Operator error	Wrong corrections or deletion	Audit history, backups, role permissions, soft-delete where appropriate.
Supply-chain dependency risk	Vulnerable or incompatible package	Lockfiles, SBOM, automated vulnerability and license scans.

18.2 Security Requirements

ID	Requirement
SEC-001	Encrypt backups and sensitive local secrets using well-reviewed libraries and OS key storage.
SEC-002	Do not store cloud provider API keys in plaintext or frontend bundles.
SEC-003	Sign Windows installers and update artifacts before public release.
SEC-004	Run OCR/parsers with least privilege and restricted file access.

ID	Requirement
SEC-005	Implement structured audit logs for login, export, delete, restore, AI request, and connector actions.
SEC-006	Provide privacy mode with telemetry disabled by default.
SEC-007	Use TLS and certificate validation for all cloud communication.
SEC-008	Perform dependency, secret, and license scans in CI.
SEC-009	Publish SECURITY.md and responsible disclosure process.
SEC-010	Complete external penetration test before enterprise launch.

18.3 Data Retention

- Local data remains until the user deletes it or the workspace retention policy acts.
- Cloud gateway should not retain raw analysis payloads by default; retain only billing/security metadata where feasible.
- If support diagnostics require uploaded files, obtain explicit consent, define expiry, and display deletion status.
- AI chat history should be locally deletable and exportable.
- Implement jurisdiction-specific data processing terms before selling into regulated markets.

19. Non-Functional Requirements

ID	Area	Target
NFR-PERF-001	App cold start	≤ 8 seconds on recommended hardware; ≤ 15 seconds on minimum supported hardware.
NFR-PERF-002	Dashboard load	≤ 3 seconds for 50,000 invoice records after indexes/aggregates.
NFR-PERF-003	OCR throughput	Median ≤ 8 seconds per clean single-page invoice on recommended CPU.
NFR-PERF-004	Idle memory	Target ≤ 600 MB desktop app excluding active OCR worker.
NFR-PERF-005	OCR memory	Target peak ≤ 1.5 GB per worker; one worker by default.
NFR-REL-001	Crash recovery	No loss of approved data; interrupted jobs resume or fail visibly.
NFR-REL-002	Backup restore	100% reproducible in automated round-trip tests.
NFR-PORT-001	Data export	All customer structured data exportable without cloud service.
NFR-ACC-001	Accessibility	Keyboard navigation, visible focus, semantic labels, contrast, screen-reader basics.
NFR-OBS-001	Diagnostics	Local logs redact financial values by default; support bundle is user-reviewed.
NFR-LOC-001	Localization	All UI strings externalized from first release; English is initial language.
NFR-UPD-001	Updates	Automatic update optional; offline installer path always available.

19.1 Hardware Support Policy

Tier	Profile	Expectation
Minimum supported	64-bit Windows 10/11, 8 GB RAM, dual/quad-core x64 CPU, 5 GB free SSD/HDD. No discrete GPU required.	Core app and single-worker OCR; slower batches.
Recommended	Windows 11, modern i5/Ryzen 5 or better, 16 GB RAM, SSD, 10+ GB free.	Smooth batch processing and analytics.

Tier	Profile	Expectation
4 GB legacy mode	Not guaranteed for production. A reduced import/report viewer prototype may be benchmarked.	No promise of local OCR performance or support.
Local LLM mode	Future advanced tier only; hardware profile published per model.	Not part of MVP.

20. Monetization and Entitlements

Plan	Features	Commercial role
Free Offline	One local company, document import, English OCR allowance or unlimited local processing subject to device, review, deterministic alerts, core dashboards, standard reports, CSV/PDF export.	Build trust and distribution.
Pro AI	Cloud AI CFO, Ask AI on reports, advanced narrative reports, scenarios, higher automation limits, priority support.	Monthly/annual subscription.
Business	Multiple companies/users, connector packs, policy controls, centralized license/admin, advanced audit exports.	Per company or seat plus usage.
Enterprise/Private	Private model endpoint, on-prem/private cloud gateway, SSO, custom retention, security review, dedicated connectors.	Annual contract.

20.1 Pricing Principles

- Do not charge users per visible “question” in a confusing way; use transparent fair-use limits and plan allowances.
- Price against measurable value: hours saved, errors prevented, faster collections, and decision visibility.
- Keep access to local customer data independent from subscription status.
- When premium expires, saved AI reports remain readable locally.
- Do not subsidize unlimited expensive model usage in low-priced plans; implement budgets and graceful limits.

21. Product Metrics and Pilot Success

Metric group	Definition
North-star	Verified documents processed with meaningful time saved per active company.
Activation	Company created + 20 documents imported + first report generated within 7 days.
OCR value	Median review time per invoice and percentage of fields accepted without correction.
Operational value	Alerts acknowledged, duplicates prevented, overdue items surfaced.
Retention	Companies active in 4 of first 6 weeks.
Premium conversion	Active free companies using an AI preflight and upgrading after demonstrated need.
Trust	AI answers opened with supporting data; correction/flag rate; user confidence survey.
Reliability	Crash-free sessions, successful backup rate, connector error rate.
Economics	Cloud AI cost as percentage of subscription revenue.

21.1 Pilot Exit Criteria

- At least 5 pilot companies complete 30 days of real use.
- At least 2,500 supported invoices processed across the cohort.
- Median operator time per invoice reduced by at least 40% versus baseline for supported layouts.
- No unrecoverable data loss.
- False auto-approval of incorrect invoice totals below threshold.
- At least 3 pilot companies state willingness to pay for a defined plan.
- At least one recurring decision-support workflow is used weekly.
- Founders can explain the top three differentiators versus existing accounting software using measured evidence.

22. Quality and Testing Strategy

Test type	Coverage
Unit tests	Money/date normalization, rules, scores, report calculations, redaction, schema validation.
Golden document tests	Invoice fixture → expected extracted JSON; versioned by engine.
Property tests	Totals, currency precision, duplicate logic, idempotency.
Integration tests	Desktop <-> worker <-> database; cloud gateway provider adapters; connector mocks.
End-to-end tests	Setup, import, review, report, backup/restore, premium consent flow.
Performance tests	50k records, 1k-document batch, low-end hardware benchmarks.
Security tests	Malicious PDFs, path traversal, prompt injection, secrets, authorization, update validation.
Migration tests	Upgrade from each supported database schema version.
Usability tests	Finance operator task completion and error rate.
AI evaluation	Grounded numeric QA, citation validity, refusal/guardrail cases, provider regression suite.

22.1 Definition of Done for Every Feature

- Requirement ID linked in issue/PR.
- Acceptance criteria pass.
- Automated tests added or reason documented.
- Error states and offline behavior implemented.
- Accessibility checked.
- Audit/security impact reviewed.
- Documentation updated.
- No new unreviewed dependency or license.
- Telemetry and data-flow changes documented.
- Product owner demo completed for user-facing features.

23. Delivery Roadmap



Figure 4 — Phased delivery roadmap

Phase	Indicative duration	Exit deliverables
P0 — Discovery & license gate	2–4 weeks	Choose Option A/B, interview users, collect invoice corpus, benchmark Frappe Books and OCR stack, define pilot connector.
P1 — Offline invoice MVP	6–10 weeks	Desktop workspace, import, local store, OCR worker, review queue, invoice register, backup.
P2 — Intelligence & reports	6–8 weeks	Rules, dashboard, supplier trends, payment priority, management pack, custom reports.
P3 — Cloud AI CFO	4–8 weeks	Gateway, consent, provider adapter, grounded answers, Ask AI, cost controls, subscriptions.
P4 — Integrations & pilots	6–12 weeks	Chosen connector, onboarding tools, telemetry opt-in, pilot support, quality improvements.
P5 — Expansion	Ongoing	Languages, vertical packs, additional connectors, enterprise deployment, private AI endpoints.

Schedule caution: Durations are planning ranges, not commitments. Codex can accelerate implementation, but user research, accounting validation, OCR corpus creation, security review, and pilot learning remain human responsibilities.

24. OpenAI Codex Implementation Plan

24.1 Rules for Codex

- Work only on one milestone at a time.
- Before coding, inspect the repository and write a short implementation plan plus risks.
- Never introduce a new dependency without recording license, purpose, alternatives, and version.
- Never implement blockchain, autonomous payments, or full accounting unless a later approved PRD changes scope.
- Use test fixtures with synthetic data; do not commit real customer invoices.
- Prefer small, reviewable pull requests.
- Write Architecture Decision Records for irreversible decisions.
- Keep domain logic independent of UI, OCR vendor, and cloud model provider.
- Run all tests and provide verification output before marking a work package complete.
- Do not claim compliance, accounting correctness, or security certification without evidence.

24.2 Work Packages

WP	Name	Deliverable
WP-00	Repository and governance	Monorepo, linting, test runners, CI, SBOM, notices, ADR template, threat-model skeleton.
WP-01	Desktop foundation	Tauri/React workspace shell, routing, local settings, secure capability allowlist.
WP-02	Database/domain	SQLite migrations, core entities, money/date types, repositories, audit events.
WP-03	Document store	Managed file import, hashes, metadata, duplicate detection, safe paths.
WP-04	OCR worker	Python service, job protocol, preprocessing, engine adapter, structured result.
WP-05	Review UI	Side-by-side viewer, confidence fields, corrections, approval state machine.
WP-06	Rules and dashboard	Rule engine, events, queries, ECharts dashboard, filters.

WP	Name	Deliverable
WP-07	Reports	Templates, custom builder, PDF/CSV exports, management pack.
WP-08	Backup/restore	Encrypted archive, verification, round-trip tests.
WP-09	Cloud gateway	Auth, tenant policy, provider interface, structured output, cost metadata.
WP-10	AI advisor	Consent preflight, context builder, citations, answer UI, safety tests.
WP-11	Connector SDK	Interfaces, CSV connector, one pilot connector.
WP-12	Packaging/release	Signed installers, updater, crash-safe migrations, offline installer.
WP-13	Pilot readiness	Performance, support bundle, user docs, onboarding, evaluation dashboards.

24.3 Ready-to-Paste Master Prompt for Codex

```
You are implementing the AI Financial Operating System described in
AI_Financial_Operating_System_PRD_Blueprint_v1.0.md.

Treat the PRD as the product source of truth. Start with WP-00 only.

Before making changes:
1. Inspect the repository and current files.
2. Summarize the intended architecture and identify conflicts with the PRD.
3. Propose a small implementation plan for WP-00.
4. List every new dependency with its license and purpose.

Implementation constraints:
- Offline-first Windows desktop.
- React + TypeScript frontend in a Tauri shell.
- SQLite operational database; DuckDB only where analytical queries justify it.
- Python document worker behind a narrow versioned job API.
- No local LLM, blockchain, autonomous payments, or full ERP in MVP.
- Financial calculations must be deterministic and tested.
- AI provider integration must remain behind an internal adapter.
- Never commit real financial data or secrets.
- Add tests, documentation, ADRs, and verification output.

Complete only WP-00, then stop and provide:
- files changed,
- commands run and results,
- open risks,
- recommended next work package.
```

24.4 Prompt for Each Subsequent Work Package

```
Implement work package <WP-ID> from the PRD.
```

First inspect the current repository and prior ADRs. Then provide:
- requirement IDs covered,
- proposed files/modules,
- data migrations,
- dependency/license changes,
- security/privacy impact,
- test plan,
- rollback/migration risk.

After approval in the same task, implement the smallest complete vertical slice.
Run lint, type checks, unit tests, integration tests, and relevant end-to-end tests.
Do not proceed to the next work package.

25. Milestone Acceptance Criteria

Milestone	Acceptance
M1 — Foundation	Installs on clean Windows VM; creates workspace; no internet required; CI green; licenses documented.
M2 — Invoice flow	Imports batch; hashes sources; local OCR; review/correction; approves record; survives restart.
M3 — Intelligence	Rules trigger reproducibly; dashboard reconciles to records; management report exports offline.
M4 — Premium AI	Consent works; only selected context is sent; numeric claims cite records; budget limits enforced.
M5 — Pilot	Backup/restore verified; pilot connector stable; performance targets met; user docs and support process ready.

26. Competitive Positioning and USP

Competitor category	Strength	AIFOS differentiation
Tally / QuickBooks / Zoho / Xero	System of record, bookkeeping, tax/accounting workflows	Cross-document intelligence, offline OCR review, explainable SME decision layer, works alongside existing software.
ERPNext / Odoo	Broad ERP modules and workflows	Narrower, simpler desktop intelligence, lower adoption burden, privacy-first optional cloud.
Cloud AI accounting startups	Automation and AI-native workflows	Offline core, customer-controlled data boundary, low-connectivity use, connector-neutral.
Generic OCR tools	Extract text/fields	Finance validation, entity history, rules, reports, and decision context.
Spreadsheets/manual processes	Flexible and familiar	Traceability, repeatability, exceptions, search, and reduced clerical work.

26.1 Defensible USP Candidates

- Offline-first invoice intelligence for SMEs using ordinary Windows PCs.
- Explainable rules plus optional AI—not a black-box chatbot.
- Industry packs for one vertical, including supplier/item terminology and relevant reports.
- Works above existing accounting systems instead of forcing migration.
- Human-verifiable OCR with source bounding boxes and correction history.
- Custom management packs plus an “Ask AI” button grounded in the exact report.

- Privacy preflight showing what leaves the device.

USP warning: “Uses AI,” “uses blockchain,” or “has dashboards” are not defensible USPs. The product must prove a measurable workflow improvement for a clearly defined vertical.

27. Risks, Assumptions, and Mitigations

ID	Risk	Severity	Mitigation
R-01	OCR accuracy varies widely	High	Start with supported document policy, human review, supplier templates, and benchmark corpus.
R-02	Users expect full accounting	High	Position as intelligence layer; integrate/export; clear scope and onboarding.
R-03	Licensing mistake in open-source fork	High	License gate, SBOM, counsel, clean-room boundaries, notices.
R-04	Cloud AI privacy concerns	High	Explicit consent, minimization, local-first design, provider contracts, enterprise private option.
R-05	AI gives confident wrong advice	High	Deterministic calculations, citations, assumptions, evaluation suite, human approval.
R-06	Low-end performance	Medium/High	No local LLM, single OCR worker, benchmarks, queueing, lazy loading.
R-07	ERP connector fragmentation	Medium/High	Connector SDK, one pilot connector, version matrix, idempotency.
R-08	Freemium cloud costs	Medium	Usage budgets, model routing, caching, paid entitlements.
R-09	Generic product lacks differentiation	High	Choose a vertical, measure time/cost savings, build industry pack.
R-10	Tax/legal localization complexity	High	Limit claims, use jurisdiction packs, professional review, no auto-filing in MVP.
R-11	Founder over-scopes product	High	Enforce work packages and explicit Won’t list.
R-12	Data loss undermines trust	Critical	Transactional DB, backups, migration tests, crash recovery, release quality gates.

28. Future Blockchain Module — Deferred

Blockchain is not part of the MVP. It should be reconsidered only when multiple independent organizations need to verify a shared event and no trusted central operator is acceptable. Most invoice intelligence inside one company is better served by a normal database and cryptographic hashes.

28.1 Possible Later Uses

- Anchor document hashes or audit-pack manifests to an external ledger for independent timestamp verification.
- Inter-company invoice authenticity network where supplier and buyer both sign a shared invoice identity.
- Regulated escrow or contract workflows in a separate product, subject to financial/legal licensing.
- Supply-chain event verification involving multiple parties.

28.2 Preconditions

- Validated customer problem that cannot be solved by signed records and a trusted service.
- Legal recognition and dispute process defined.
- Privacy design avoids putting confidential invoice data on a public ledger.
- Fees, key recovery, governance, and operational failure modes are acceptable.
- User experience hides blockchain complexity.

29. Decisions Required Before Production Coding

ID	Decision	Question	Owner
D-01	License/business model	Proprietary permissive core or AGPL open-core Frappe Books fork?	Founder + legal
D-02	Pilot vertical	Handicrafts, manufacturing, trading, or another narrow segment?	Founder
D-03	Pilot connector	Tally, CSV-only, ERPNext, or another system?	Founder + pilot customers
D-04	Supported invoice policy	English printed only; file and quality limits.	Product + OCR lead
D-05	Primary cloud AI provider	Select after benchmark, data terms, cost, and structured-output tests.	Engineering + legal
D-06	Free-plan limits	Company count, document volume, support, and premium trigger.	Founder
D-07	Data hosting region	Required only for cloud accounts/gateway metadata.	Legal + operations
D-08	Minimum Windows versions	Confirm from pilot hardware survey.	Engineering
D-09	Accounting validation owner	Qualified accountant/advisor to review calculations and reports.	Founder
D-10	Product name and trademark	Perform search before public branding.	Founder + legal

Appendix A — Official Source and License References

ID	Project	Official link	License / note
OS-01	Frappe Books	https://github.com/frappe/books	AGPL-3.0; offline Electron/Vue/SQLite accounting app.
OS-02	ERPNext	https://github.com/frappe/erpnext	GPL-3.0 ERP.
OS-03	Frappe Framework	https://github.com/frappe/frappe	MIT full-stack framework.
OS-04	Odoo	https://github.com/odoo/odoo	Community repository; LGPL-3.0 license file.
OS-05	Tauri	https://github.com/tauri-apps/tauri	MIT or Apache-2.0 where applicable.
OS-06	PaddleOCR	https://github.com/PaddlePaddle/PaddleOCR	Apache-2.0; OCR/document toolkit.
OS-07	Tesseract	https://github.com/tesseract-ocr/tesseract	Apache-2.0 OCR engine.
OS-08	Docling	https://github.com/docling-project/docling	MIT codebase; individual model licenses vary.
OS-09	SQLite	https://sqlite.org/	Public domain; embedded

ID	Project	Official link	License / note
		about.html	transactional database.
OS-10	DuckDB	https://github.com/duckdb/duckdb	MIT analytical database.
OS-11	Apache ECharts	https://github.com/apache/echarts	Apache-2.0 charting library.
OS-12	Ollama	https://github.com/ollama/ollama	Recheck application/runtime license and each model license before bundling.
OS-13	Qwen3	https://github.com/QwenLM/Qwen3	Official repository states open-weight models under Apache-2.0; verify selected artifact.
OS-14	LangChain	https://github.com/langchain-ai/langchain	MIT.
OS-15	LlamaIndex	https://github.com/run-llama/llama_index	MIT; optional dependencies require review.
OS-16	Qdrant	https://github.com/qdrant/qdrant	Apache-2.0.
OS-17	Chroma	https://github.com/chroma-core/chroma	Apache-2.0.
Verification date: Repository descriptions and license labels were checked on 3 August 2026. Recheck at the exact commit/tag selected for production. This appendix is not legal advice.			

Appendix B — Initial Feature Priority Matrix

Priority	Features
Must	Workspace, import, hashing, English OCR, review, duplicate/validation rules, dashboard, offline reports, backup/restore.
Should	Custom reports, price trends, payment priority scoring, saved templates, one ERP connector, premium AI preflight and grounded Q&A.
Could	Additional languages, mobile capture, multiple cloud providers, private model endpoint, vector search, advanced anomaly models.
Won't now	Blockchain, crypto, autonomous payments, full ERP/accounting replacement, payroll, all languages, guaranteed handwriting.

Appendix C — Glossary

Term	Definition
AIFOS	AI Financial Operating System, the working product name.
OCR	Optical character recognition.
ERP	Enterprise resource planning system.
Staging	Validated records used for intelligence before/without posting to a formal ledger.
Rule engine	Deterministic logic that produces repeatable alerts/calculations.
Grounding	Restricting an AI answer to supplied data and citations.
Context minimization	Sending only information necessary for a cloud AI task.
Idempotency	Repeating an operation without creating duplicate effects.
SBOM	Software bill of materials listing included dependencies/components.
AGPL	Strong copyleft software license with network interaction provisions; obtain legal advice.

Term	Definition
ADR	Architecture Decision Record.
Golden fixture	Document and expected output used for regression testing.

Appendix D — Founder Review Checklist

☐ Confirm the product is an intelligence layer, not a complete accounting replacement.

☐ Select the license strategy and get written legal guidance.

☐ Choose one pilot vertical and recruit at least five design partners.

☐ Collect representative, permissioned invoice samples.

☐ Measure the current manual workflow before building.

☐ Choose one connector or use CSV-only for first pilot.

☐ Appoint a qualified accounting reviewer.

☐ Approve free versus premium boundaries.

☐ Approve privacy preflight wording and data retention.

☐ Give Codex only WP-00 first; review before moving to WP-01.

Final Recommendation

Proceed, but proceed narrowly. The best first product is not “an AI ERP” and not “blockchain accounting.” It is an offline invoice-intelligence and management-reporting layer for one type of SME, with optional cloud AI for deeper explanations and scenarios. Build the source-of-truth calculations locally, make every result auditable, and earn the right to expand through measured pilot value.